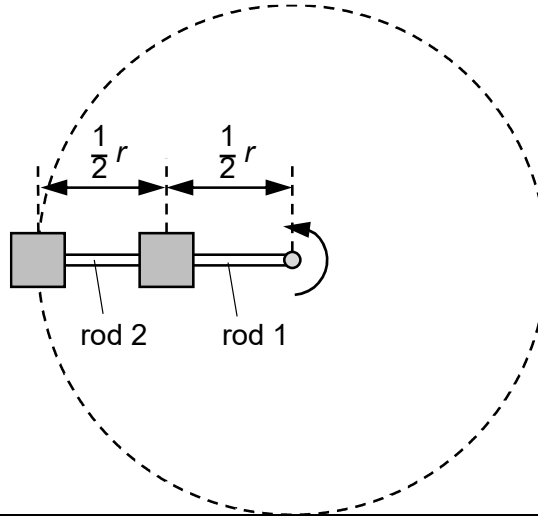


10

Two identical blocks connected by light rigid rods are pivoted at one end to move in a circular path at uniform angular velocity on a smooth horizontal surface, as shown.

**[Turn over**

What is the ratio of $\frac{\text{tension in rod 1}}{\text{tension in rod 2}}$?							
A	0.50	B	1.0	C	1.5	D	2.0

Answer: C

$$F_2 = mr\omega^2 \dots\dots (1)$$

$$F_1 - F_2 = m\left(\frac{1}{2}r\right)\omega^2 \dots\dots (2)$$

$$(1) + (2): F_1 = \frac{3}{2}mr\omega^2$$

$$\text{so, ratio} = \frac{\frac{3}{2}mr\omega^2}{mr\omega^2} = 1.5$$